

3.20 Binary turtle graphics – version 6

In computer graphics, turtle graphics are vector graphics using a relative cursor (the "turtle") upon a Cartesian plane. The turtle has three attributes: a location, an orientation (or direction), and a pen. The pen, too, has attributes: color, on/off (or up/down) state [2].

The turtle moves with commands that are relative to its own position, such as "move forward 10 spaces" and "turn left 90 degrees". The pen carried by the turtle can also be controlled, by enabling it or setting its color.

Your task is to write a program, which translates binary encoded turtle commands to a raster image in a BMP file [1].

Turtle commands

The length of all turtle commands is 16 or 32 bits. The first two bits define one of four commands (set position, set direction, move, set state). Unused bits in all commands are marked by the – character. They should not be taken into account when the command is decoded.

Set position command

The *set position* command sets the new coordinates of the turtle. It consists of two words. The first word defines the command (bits 15-14) and Y (bits y5-y0) coordinate of the new position. The second word contains the X (bits x9-x0) coordinate. The point (0,0) is located in the bottom left corner of the image.

Table 1. The first word of the *set position* command.

bit no. 15	14	13	12	11	10	9	8
1	0	y5	y4	y3	y2	y1	y0
bit no. 7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-

Table 2. The second word of the *set position* command.

bit no. 15	14	13	12	11	10	9	8
x9	x8	x7	x6	x5	x4	x3	x2
bit no. 7	6	5	4	3	2	1	0
x1	x0	-	-	-	-	-	-

Set direction command

The *set direction* command sets the direction in which the turtle will move, when a move command is issued. The direction is defined by the d1, d0 bits.

Table 3. The *set direction* command.

bit no. 15	14	13	12	11	10	9	8
1	1	-	-	d1	d0	-	-
bit no. 7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-

Table 4. The description of the d1,d0 bits.

Bits d1,d0	Turtle direction
00	up
01	left
10	down
11	right

Move command

The *move* command moves the turtle in direction specified by the d1-d0 bits. The movement distance is defined by the m9-m0 bits. If the destination point is located beyond the drawing area the turtle should stop at the edge of the drawing. It can't leave the drawing area. The turtle leaves a visible trail when the pen is lowered (bit ud). The color of the trail is defined by the r3-r0, g3-g0, b3-b0 bits.

Table 5. The *move* command.

bit no. 15	14	13	12	11	10	9	8
0	0	m9	m8	m7	m6	m5	m4
bit no. 7	6	5	4	3	2	1	0
m3	m2	m1	m0	-	-	-	-

Set pen state command

The *pen state* command defines whether the pen is raised or lowered (bit ud) and the color of the trail. Bits c2-c0 select one of the predefined colors from the color table.

Table 6. The *pen state* command.

bit no. 15	14	13	12	11	10	9	8
0	1	-	ud	-	c2	c1	c0
bit no. 7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-

Table 7. The description of the ud bit.

ud bit	Pen state
0	pen lowered (down)
1	pen raised (up)

Table 8. Color table.

bits c2,c1,c0	Pen state
000	black
001	purple
010	cyan
011	yellow
100	blue
101	green
110	red
111	white

Input

- binary file containing 16/32-bits turtle commands
- file name: “input.bin”

Output

- BMP file containing the generated image:
 - Sub format: 24 bits RGB – no compression,
 - Image size: 600x50 px,
- file name: “output.bmp”

References:

- [1] BMP file format – see section 4.2
- [2] “**Turtle graphics**”, https://en.wikipedia.org/wiki/Turtle_graphics
- [3] Example program for bmp reading/writing,
http://galera.ii.pw.edu.pl/~zsz/ecoar/bmp/bmp_riscv.zip